DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY LUCKNOW



STUDY, EVALUATION SCHEME & SYLLABUS

For

B. Voc.
Banking Finance Services and Insurance (FS)
Branch Code: 108

Based on
AICTE Model Curriculum

(EFFECTIVE FROM THE SESSION: 2019-20)

EVALUATION SCHEME
Banking Finance Services and Insurance (BFSI)

BFSI Level 5 SEMESTER- I											
S. No.	Subject Code	Subject	Total Teaching/ Training Hours	Evaluation Scheme				End Semester		Total	Credit
				CT	TA	AT	Total	TE	PE		
1	BFSV511	Banking & Micro Finance - I	30	10	5	5	20	30		50	2
2	BFSV512	Indian Securities Markets - I	30	10	5	5	20	30		50	2
3	BFSV513	Micro Finance Operations - I	30	10	5	5	20	30		50	2
4	BFSV514	Mutual Fund Operations - I	30	10	5	5	20	30		50	2
5	BFSP511	Vocational Practical –I	30	-	-	-	20	-	30	50	1
6	BFSP512	Vocational Practical –II	30	-	-	-	20	-	30	50	1
7	BFSP513	Language Lab	30	-	-	-	20	-	30	50	2
8	BFST511	Insurance Agent (BSC/Q3801)						Any one Training 400hrs/ 8 weeks		150	12
	BFST512	Business Correspondent / Business Facilitator(BSC/Q8401)									
	BFST513	Debt Recovery Agent (BSC/Q2303)									
Total			610	-	-	-	-	-	-	500	24

BFSI Level 5 SEMESTER- II											
S. No.	Subject Code	Subject	Total Teaching/ Training Hours	Evaluation Scheme				End Semester		Total	Credit
				CT	TA	AT	Total	TE	PE		
1	BFSV521	Banking & Micro Finance – II	30	10	5	5	20	30		50	2
2	BFSV522	Indian Securities Markets - II	30	10	5	5	20	30		50	2
3	BFSV523	Micro Finance Operations - II	30	10	5	5	20	30		50	2
4	BFSV524	Mutual Fund Operations - II	30	10	5	5	20	30		50	2
5	BFSP521	Vocational Practical –III	30	-	-	-	20	-	30	50	1
6	BFSP522	Vocational Practical –IV	30	-	-	-	20	-	30	50	1
7	BFSP523	IT Tool Lab	30	-	-	-	20	-	30	50	2
8	BFST521	Mutual Fund Agent (BSC/Q3802)						Any one Training 400 hrs/ 6 weeks		150	12
	BFST522	Small and Medium Enterprise (SME) Officer (BSC/Q2302)									
Total			610	-	-	-	-	-	-	500	24

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY LUCKNOW

STUDY, EVALUATION SCHEME & SYLLABUS

For

B. Voc.

Electronics Manufacturing Services (EM)
Branch Code:111

Based on

AICTE Model Curriculum

(EFFECTIVE FROM THE SESSION: 2019-20)

EVALUATION SCHEME
Electronics Manufacturing Services

NSFQ Level 5 SEMESTER- I											
S. No.	Subject Code	Subject	Total Teaching/ Training Hours	Evaluation Scheme				End Semester		Total	Credit
				CT	TA	AT	Total	TE	PE		
1	BEMV511	Electronic Measurement and Instrumentation –I	30	10	5	5	20	30		50	2
2	BEMV512	Identification of Components, Tools, SOP & Work Instructions-I	30	10	5	5	20	30		50	2
3	BEMV513	Tools, Equipment & Safety Measures –I	30	10	5	5	20	30		50	2
4	BEMV514	Soldering & De-Soldering of Components –I	30	10	5	5	20	30		50	2
6	BEMP511	Identification of Components, Tools, Equipment and its working –Lab	30				20		30	50	1
7	BEMP512	Electronic Measurement and Instrumentation -I –Lab	30				20		30	50	1
8	BEMP513	Language Lab	30				20		30	50	2
8	BEMT511	Embedded Software Engineer (ELE/Q1501)	Any one Training 400 hrs/ 8 weeks							150	12
	BEMT512	Security System Service Engineer (ELE/Q4610)									
	BEMT513	Systems Analyst (ELE/Q8701)									
	BEMV511	Embedded Software Engineer (ELE/Q1501)									
Total			610							500	24

NSFQ Level 5 SEMESTER- II											
S. No.	Subject Code	Subject	Total Teaching/ Training Hours	Evaluation Scheme				End Semester		Total	Credit
				CT	TA	AT	Total	TE	PE		
1	BEMV521	Electronic Measurement and Instrumentation –II	30	10	5	5	20	30		50	2
2	BEMV522	Identification of Components, Tools, SOP & Work Instructions-II	30	10	5	5	20	30		50	2
3	BEMV523	Tools, Equipment & Safety Measures –II	30	10	5	5	20	30		50	2
4	BEMV524	Soldering & De-Soldering of Components & Emergency actions II	30	10	5	5	20	30		50	2
5	BEMP521	Soldering & De-Soldering of Components-Lab	30				20		30	50	1
6	BEMP522	Electronic Measurement and Instrumentation -II (Lab)	30				20		30	50	1
7	BEMP523	IT Tool Lab	30				20		30	50	2
8	BEMT521	Smartphone Assembly Inspector (ELE/Q4001)	Any one Training 400 hrs/ 8 weeks							150	12
	BEMT522	Business Development Executive (ELE/Q1101									
Total			610							500	24

NSFQ Level 6 SEMESTER- III											
S. No.	Subject Code	Subject	Total Teaching/ Training Hours	Evaluation Scheme				End Semester		Total	Credit
				CT	TA	AT	Total	TE	PE		
1	BEMV631	Fault analysis & Repairs	30	10	5	5	20	30		50	2
2	BEMV632	Good Manufacturing Concept & Practices – I	30	10	5	5	20	30		50	2
3	BEMV633	Electronics Devices Circuit –I	30	10	5	5	20	30		50	2
4	BEMV634	Electronics System Packaging and Manufacturing	30	10	5	5	20	30		50	2
5	BKVH631	Human Values and Professional Ethics	30	10	5	5	20	30		50	2
6	BEMP631	Electronics Devices Circuit –I Lab	30				20		30	50	1
7	BEMP632	Fault analysis & Repairs – Lab +	30				20		30	50	1
8	BEMT631	Field Engineer RACW (ELE/Q3105)					Any one Training 400 hrs/ 8 weeks			150	12
	BEMT632	Security System Service Engineer (ELE/Q4610)									
	BEMT633	Pre-Sales Solar Technical Support Engineer (ELE/Q5602)									
Total			610							500	24

NSFQ Level 6 SEMESTER- IV											
S. No.	Subject Code	Subject	Total Teaching/ Training Hours	Evaluation Scheme				End Semester		Total	Credit
				CT	TA	AT	Total	TE	PE		
1	BEMV641	Good Manufacturing Concepts Practices–II	30	10	5	5	20	30		50	2
2	BEMV642	Manufacturing & Quality Norms	30	10	5	5	20	30		50	2
3	BEMV643	Good Manufacturing Concepts & Practices–III	30	10	5	5	20	30		50	2
4	BEMV644	Electronics Devices Circuit –II	30	10	5	5	20	30		50	2
5	BKVE641	Environment and Ecology	30	10	5	5	20	30		50	2
6	BEMP641	Electronics Devices Circuit –II Lab	30				20		30	50	1
7	BEMP642	Manufacturing Practices	30				20		30	50	1
8	BEMT641	Purchase Executive (ELE/Q5701)					Any one Training 400 hrs/ 8 week			150	12
	BEMT642	Quality Engineer (ELE/Q7901									
Total			610							500	24

NSFQ Level 7 SEMESTER- V											
S. No.	Subject Code	Subject	Total Teaching/ Training Hours	Evaluation Scheme				End Semester		Total	Credit
				CT	TA	AT	Total	TE	PE		
1	BEMV751	Valuation & Storage	30	10	5	5	20	30		50	2
2	BEMV752	Shelf Life, Ware House Operations Management & Material Transactions	30	10	5	5	20	30		50	2
3	BEMV753	Industrial Electronics Product Design	30	10	5	5	20	30		50	2
4	BEMV754	Pre-Production Activities	30	10	5	5	20	30		50	2
5	BKVH751	Constitution of India, Law and Engineering	30	10	5	5	20	30		50	2
6	BEMP751	Pre-Production Activities-Lab	30				20		30	50	1
7	BEMP752	Valuation & Storage-Lab	30				20		30	50	1
8	BEMT751	Product Engineer (ELE/Q4201)					Any one Training 400 hrs/ 8 weeks			150	12
	BEMT752	Incoming QC Technician (ELE/Q4401)									
	BEMT753	Assembly Supervisor (ELE/Q6305)									
Total			610							500	24

NSFQ Level 7 SEMESTER- VI											
S. No.	Subject Code	Subject	Total Teaching/ Training Hours	Evaluation Scheme				End Semester		Total	Credit
				CT	TA	AT	Total	TE	PE		
1	BEMV761	Entrepreneurship/Accounting/Management	30	10	5	5	20	30		50	2
2	BEMV762	Trouble shooting and Maintenance of Electronics Equipment	30	10	5	5	20	30		50	2
3	BKVH761	Indian Tradition, Culture and Society	30	10	5	5	20	30		50	2
4	BEMP761	Major Project	180						150	150	6
5	BEMT761	FPGA Design Engineer (ELE/Q8201)					Any one Training 400 hrs/ 8 weeks			200	12
	BEMT762	Sales Executive-Consumer Electronics (ELE/Q3201)									
Total			670							500	24

Detailed Curriculum
Level 5 (Semester I)

Electronic Measurements and Instrumentation-I

Unit, dimensions and standards

Scientific notations and metric prefixes, SI electrical units, SI temperature scales, Other unit systems, dimension and standards. Measurement Errors: Gross error, systematic error, absolute error and relative error, accuracy, precision, resolution and significant figures, Measurement error combination, basics of statistical analysis. PMMC instrument, galvanometer, DC ammeter, DC voltmeter, series ohm meter Transistor voltmeter circuits, AC electronic voltmeter, current measurement with electronic instruments, probes Digital voltmeter systems, digital multi-meters, digital frequency meter system.

Electromagnetic Effects

Permanent magnets and Electromagnets, their construction and uses, Polarities of an electromagnet and rules for finding them, Faraday's Laws of Electromagnetic Induction, Dynamically induced e.m.f., its magnitude and induction, inductance and its unit. Mutually induced e.m.f., its magnitude and direction, Energy stored in an inductance. Force acting on a current carrying conductor in magnetic field, its magnitude and direction, Principles and construction of dynamo.

A.C Circuits

Generation of A.C. voltage, its generation and wave shape. Cycle, frequency, peak value R.M.S. value, form factor, crest factor, Phase difference, power and power factor, A.C. Series Circuits with (i) resistance and inductance (ii) resistance and capacitance and (iii) resistance inductance and capacitance, Q factor of R.L.C. series circuits.

Identification of Components, Tools, SOP & Work Instructions-I

1. Main components & modules/ sub-assemblies of electronic equipment
    - Control Panel (System Controller)
    - Keypads
    - Door and Window Contacts
    - Motion Detectors
    - Glass Break Detection
    - Smoke Detectors
    - Heat Sensors
    - Carbon Monoxide Detectors
    - Water Detectors (or Water Bug)
    - Temperature Sensors
    - Capacitance switches / control push buttons & rotary switches

2. Introduction to Basic Electricity

Current Electricity

Definition of Resistance, Voltage, Current, Power, Energy and their units, Relation between electrical, mechanical and thermal units, Temperature variation of resistance, Difference between AC and DC voltage and current.

D.C. Circuits
Ohm's Law, Series – parallel resistance circuits, calculation of equivalent resistance, Kirchhoff's Laws and their applications.

Electric Cells
Primary cell, wet cell, dry cell, battery, Li-ion battery, series and parallel connections of cells, Secondary cells, Lead Acid Cell, Discharging and recharging of cells, preparation of electrolyte, care and maintenance of secondary cells.

Lighting Effects of Current
Lighting effect of electric current, filaments used in lamps, and Tubelight, LED, their working and applications.

Capacitors
Capacitor and its capacity, Concept of charging and Discharging of capacitors, Types of Capacitors and their use in circuits, Series and parallel connection of capacitors, Energy stored in a capacitor.

3. **Introduction and Identification of Electronics Components**
   - Electronic controls in a common way
   - Diode, Transistors, Op-Amp
   - Number systems and Boolean Algebra
   - Logic Gates
   - Flip-Flops
   - Counters
   - Multiplexers
   - Decoders
   - LED, LCD and 7 segment display
   - RAM and ROM
4. Concept of Amplification factor, Gain & Signal distortion
5. Protocols like TCP/IP for communication purpose and for digital networks & circuits.

Tools, Equipment and Safety Measures-I

1. Cables & Connectors
   - Non-Metallic Sheathed Cable
   - Un grounded & Grounded Power Supply Cable
   - Metallic Sheathed Cable
   - Multi-Conductor Cable
   - Coaxial Cable
   - Unshielded Twisted Pair Cable
   - Shielded twisted pair cable
   - Ribbon Cable
   - Armoured & Unarmoured Cable
   - Twin-Lead Cable
   - Twin axial Cable
   - Optical fiber cable
   - Connectors
1. ESD Clothing
   - What to wear, how to wear

Soldering & De-Soldering of Components-I

1. Soldering & De Soldering of Basic Components
   - Soldering Tools
   - Different types of Soldering Guns related to Temperature and wattages, types of tips
   - Solder materials and their grading
   - Soldering and De Soldering Stations and their Specifications
   - Preparing Component for Soldering
   - PCB Applications
   - Types of PCB
   - Soldering Basic Components on PCB
   - De soldering Basic Components
   - Safety precautions while Soldering & De soldering
   - Check for cold continuity of PCB
   - Identification of loose/dry solder, broken tracks on printed wire assemblies & discrete components mounted circuit boards
   - Join the broken PCB track and test
   - De soldering using Pump and wick
   - Introduction of SMD Components

Identification of Components, Tools, SOP & Work Instructions-I (Lab)

1. Identification & working of various electronic components
2. Working of testing equipment
3. Measurement using Multimeter & Clamp meter
4. Battery health check-up
5. Measure and test the voltage of given cells.
6. Verification of truth tables for AND, OR, NOT and NAND logic gates.
7. Verification of truth tables for NOR, XOR and XNOR logic gates.
8. Construction and verification of operations of half adder and full adder circuits using basic gates.
9. Study and verification of truth tables for 3 line to 8 line decoder.
10. Study and verification of truth tables for 4:1 MUX using gates
11. Study and verification of truth tables for 1:4 DEMUX using gates.
12. Construction and verification of truth tables for S-R, D and J-K flip flops.
13. Study and verification of truth table for universal shift register.
14. Study the operation of a synchronous counter.

Electronic Measurement and Instrumentation -I (Lab)

1. Study of semiconductor diode voltmeter and its use as DC average responding AC voltmeter.
2. Study of L.C.R. bridge and determination of the value of the given components.
3. Study of distortion factor meter and determination of the % distortion of the given oscillator.
4. Study of the transistor tester and determination of the parameters of the given transistors.

Level 5 (Semester II)

Electronic Measurements and Instrumentation - II

Voltmeter and ammeter methods, Wheatstone bridge, low resistance measurements, low resistance measuring instruments AC bridge theory, capacitance bridges, Inductance bridges, Q meter

CRO: CRT, wave form display, time base, dual trace oscilloscope, measurement of voltage, frequency and phase by CRO, Oscilloscope probes, Oscilloscope specifications and performance. Delay time based Oscilloscopes, Sampling Oscilloscope, DSO, DSO applications Instrument calibration: Comparison method, digital multimeters as standard instrument, calibration instrument Recorders: X-Y recorders, plotters

Identification of Components, Tools, SOP & Work Instructions-II

1 Introduction to wireless communication
2 Signal Converters
3 Tools & their Uses
- Use of tester to monitor AC Power
- Skin the electrical wires/cables using the wire stripper and cutter
- Main cable for control & electronic circuit wires
- Crimping tools and buses

4 Introduction to measuring equipment's
- Signal generator's
- CRO
- Function Generators
- Frequency Counter
- Logic analyzer
- Spectrum analyzer
- LCRQ Meter

5 Standard Operating Procedures and Work Instructions
- What is SOP and WI
- How to read & follow SOP and WI
- Overall Quality Assurance Plan

Tools, Equipment & Safety Measures-II

1. Tools & Equipment
   - Types of tools & equipment required and deployed in manufacturing, installing & servicing
   - Identification and termination process
   - General maintenance of tools/equipment and recalibration of Test equipment
   - General safety and common-sense safety
2. PPE
   - Usage & benefits of PPE

- Types & usage of various PPE
- Maintenance of PPE

2. Clean Room Environment
   - Do's and Don't
   - Shop Floor Discipline

Soldering & De-soldering components & Emergency actions

1. Introduction to SMD Components
   - Identification of 2, 3, 4 terminal SMD components
   - Soldering the SMD components on the PCB
   - Make the necessary settings on SMD soldering station to solder various ICs of different packages by choosing proper clamping tools
   - Identify various connections and the setup required for SMD soldering station
   - De solder the SMD components from the given PCB
   - Make the necessary settings on SMD soldering station to de solder various ICs of different packages by choosing proper clamping tools
   - Make a panel board using different types of switches for a given application
   - Identification of crimping tools for various IC packages
   - Reliable Soldering Practices

2. Emergency actions
   - Minimum Requirements
   - Reporting Emergencies
   - Emergency exits
   - Primary and secondary evacuation routes
   - Locations of fire extinguishers
   - Fire alarm pull stations' location
   - Assembly points
   - Medical Services

Soldering & De-soldering of components - II Lab

1. Assemble the product
2. Dis-assemble the product
3. Safety Precautions & emergency plans

Electronic Measurement-II Lab

1. Study of the following transducer (i) PT-100 trans (ii) J- type trans. (iii) K-type trans (iv) Pressure transducer
2. Measurement of phase difference and frequency using CRO (lissajous figure)
3. Measurement of low resistance Kelvin's double bridge.
4. Radio Receiver Measurements

Level 6 (Semester III)
Fault Analysis & Repairs

1. Classification of fault
2. Identification of fault
3. Rectification of fault
4. Repairing/Replacing Module
5. Analysis for the different types of equipment's
   - Smartphones
   - Air Conditioning
   - Security systems
   - Electronically controlled doors
6. Fault analysis based on hardware and software component
7. Diagnostic and Testing Methods
8. Visual Inspection
9. Earth Continuity Test
10. Insulation Resistance Test

(6.GV.02) Good Manufacturing Concept & Practices – I

- TQM (Total Quality Management) & Kaizen
- Inventory Management & Logistics in brief
- Quality assurance
- Checklist
- SWOT analysis
- Lean Manufacturing
- Muda, Mura & Muri – Toyota Production System (TPS)
- Spatial considerations & other related concepts

Electronics Devices & Circuit-I

Unit I

Energy Bands and Charge Carrier in Semiconductor: Bonding forces and energy bands in solids, Charge Carriers in Semiconductors, Carrier Concentrations, Drift Mechanism. Excess carriers in Semiconductors: Optical Absorption, Carrier Lifetime: Direct Recombination, Steady State Carrier Generation, Quasi-Fermi Level, Diffusion of carriers and Einstein relation.

UNIT II Junctions: Equilibrium Conditions, Forward and Reveres Biased Junctions; Steady State Conditions. Optoelectronic Devices: Photodiode V-I characteristic, Photodetector, Solar Cells, Light Emitting Diode. Introduction to Thyristors, PNPN diode, SCR, LASCR, DIAC, TRIAC

Electronics System Packaging and Manufacturing

Evolution and Classification of Printed Circuit Boards, Challenges in Modern PCB Design and Manufacture, PCB fabrication methodologies (SSB, DSB and multilayer board), PCB design considerations/ design rules for analog, digital and power applications, Electromagnetic

interference in electronic systems and its impact. Analysis of electronic circuit from noise emission point of view (both conducted and radiated emission) cross talk and reflection behavior of the circuit in time domain, Thermal management of electronic devices and systems.

Semiconductor Packages: Single chip packages or modules. (SCM) Commonly used packages and advanced packages; Materials in packages, Current trends in Packaging, Multichip modules (MCM)-types; System-in package (SIP); Packaging roadmaps; Hybrid circuits. Pipe and FIFOs, Shared memory, Sockets

Electronic Devices and Circuits Lab

1. Study of Lab Equipments and Components: CRO, Multimeter, and Function Generator, Power supply- Active, Passive Components and Bread Board.
2. P-N Junction diode: Characteristics of PN Junction diode - Static and dynamic resistance measurement from graph.
3. Applications of PN Junction diode: Half & Full wave rectifier- Measurement of Vrms, Vdc, and ripple factor.
4. Characteristics of Zener diode: V-I characteristics of zener diode, Graphical measurement of forward and reverse resistance.
5. Application of Zener diode: Zener diode as voltage regulator. Measurement of percentage regulation by varying load resistor.
6. Study CE configuration for NPN and PNP transistors and measurement of voltage and current gain.
7. Study CB configuration for NPN and PNP transistors and measurement of voltage and current gain.
8. Study CC configuration for NPN and PNP transistors and measurement of voltage and current gain.

Fault Analysis & Repairs - Lab

1. Categorization of faults
   - Hardware/Software, User Induced, Component Failures
   - L0 to L4 repairs
2. Testing electrical/electronic components in the product
3. Troubleshoot and repair of the faults identified in the product
   - Microphone
   - Musical Instruments (Loudspeakers)
   - Recorder (CD/ DVD Player)
4. Preventive Maintenance Services
5. Basic Occupational Safety and Precautions
6. Microphones and Loudspeakers

Level 6 (Semester IV)

Good Manufacturing Concepts & Practices – II

- Work Study Concepts
  - Method study
  - Work measurement
  - Sequencing of Operations and timing the flow steps
  - Advantages of work study

- Team Working
  - Forming
  - Storming
  - Norming
  - Performing
  - Adjourning

Manufacturing & Quality Norms

1. Manufacturing & Quality Norms- keep it differently according to all applications
   - Manpower Deployment and Operations as per Work Instructions and criticality of the process Understanding how to form each operation and practical training of operation
   - Understanding accept and reject criterion of a particular operation. Practical training of testing/checking each operation
   - Quality Norms of accept and practical training of electronic equipment's/Devices
   - Acceptance/ Rejection training of various defects

2. Manufacturing & Quality Norms – II
   - Process in packing line-Packing line Operations sequence flow and its importance
   - Quality Systems - Accept, Reject criterion of various tests at OQA
   - Training of Assembly of electronic components - Assemble, Check, test electronic components
   - Various Labels and their Importance - Understanding Labels, Scanning and its importance
   - Packing of components/devices - Various Stages of packing
   - Acceptance, Reject and sampling following QA norms - AQL level, Sampling techniques, as per QA sampling accept, reject numbers

Good Manufacturing Concepts & Practices – III

1. Good Manufacturing Concepts & Practices - II
   - Brief Introduction
   - Total Quality Management
     - ISO Standards
   - Kaizen
   - Toyota Production System

- Lean Manufacturing
  - Combination of Inventory
  - Supply Chain
- Quality and Inspection
  - 3 Sigma and 6 Sigma Orientation

Electronic Devices and Circuits –II

UNIT I MOSFET: Device structure and its operation in equilibrium, V-I characteristics. Circuits at DC, MOSFET as Amplifier and switch, Biasing in MOS amplifier circuits, small-signal operation and models, single stage MOS amplifier, MOSFET internal capacitances and high frequency model, frequency response of CS amplifier

UNIT II BJT: Review of device structure operation and V-I characteristics, BJT circuits at DC, BJT as amplifier and switch, biasing in BJT amplifier circuit, small-signal operation and models, single stage BJT amplifier, BJT internal capacitances and high frequency model, frequency response of CE amplifier.

UNIT III Feedback: The general feedback structure, properties of negative feedback, the four basic feedback topologies, the series-shunt feedback amplifier, the series-series feedback amplifier, the shunt-shunt and shunt series feedback amplifier. Oscillators: Basic principles of sinusoidal oscillators, op-amp RC oscillator circuits, LC oscillator

Electronic Devices and Circuits –II Lab

1. Characteristic of BJT: BJT in CE configuration- Graphical measurement of hparameters from input and output characteristics. Measurement of Av, AI, Ro and Ri of CE amplifier with potential divider biasing.
2. Measurement of Operational Amplifier Parameters: Common Mode Gain, Differential Mode Gain, CMRR, Slew Rate.
3. Applications of Op-amp: Op-amp as summing amplifier, Difference amplifier, Integrator and differentiator.
4. Field Effect Transistors: Single stage Common source FET amplifier –plot of gain in dB Vs frequency, Measurement of, bandwidth, input impedance, maximum signal handling capacity (MSHC) of an amplifier.
5. Oscillators: Sinusoidal Oscillators a. Wein's bridge oscillator b. phase shift oscillator.

Manufacturing Practices

1. Work study concepts
2. Team work concepts

Level 7 (Semester V)

Valuations & Storage

1. Valuation
    - Specific Item cost
    - Weighted average cost
2. Storage
    - Stacking Norms
    - Bin Cards
    - Stores Layout
    - Categorization of Materials
        - Hazardous/Non-Hazardous
        - Imported/Local
        - Assembly/ Parts
        - Consumables
        - Class A/B/C
        - Good/defective

Shelf Life, Ware House Operations Management & Material Transactions

1. Shelf Life Management
    - FIFO
    - FILO
    - LIFO
    - LILO

2. Material Transactions
    - Inward
    - Outward
    - Suspense
    - RMA (Return Material Authorization)
    - Insurance

Industrial Electronic Product Design

Development Process, Product Planning & Conceptualization, Product Architecture and Industrial Design, Product Manufacturing & Prototyping, Economic Analysis & Managing Projects. Introduction to 3-D printing and Rapid Prototyping

Pre-Production Activities

Pre-Production activities
- Layout
- Time Study & Motion Study
- Two Hand Insertion
- Non-value adding activities
- Positioning of Bins
- Line Balancing

Pre-Production Activities Lab

- Pre-Production activities
  - Two Hand Insertion
  - Positioning of Bins

- House Keeping
- 5S

Valuations & Storage Lab

- Categorization of Raw Material & Consumables
  - Hazardous/Non-Hazardous
  - Imported/Local
  - Assembly/Parts
  - Class A/B/C
  - Good/defective

- Material Transactions
  - Inward
  - Outward
  - Suspense

Level 7 (Semester II)

Accounting & Management/ Entrepreneurship

1. Introduction
Meaning and Nature of Management, Management Approaches, Processes, Managerial Skills, Tasks and Responsibilities of a Professional Manager.

2. Organizational Structure and Process
Organizational Culture and Climate, Managerial Ethos, Organization Structure & Design, and Managerial Communication.

3. Planning and Controlling
Planning Types and Process, Management by Objectives, Decision-Making Types and Models, Problem Solving Techniques, Controlling: Process and Techniques.

4. Performance Evaluation Techniques:
Introduction to Budgeting and Budgetary Control; Performance Budgeting; Classification of Budget; Standard Costing and Variance Analysis; Balanced Scorecard; Responsibility Accounting.

5. Decision Making Techniques:
Cost Volume Profit Analysis; Management Accounting for Decision Making and Control; EVA and Performance Measurement; Introduction to Activity Base Costing, Targeting Costing, Life Cycle Costing; Uniform Costing.

Or

Course Contents:

1. Entrepreneurship: Concept and Definitions; Entrepreneurship and Economic Development; Classification and Types of Entrepreneurs; Entrepreneurial Competencies; Factor Affecting Entrepreneurial Growth – Economic, Non-Economic Factors; EDP Programmes; Entrepreneurial Training; Traits/Qualities of an Entrepreneurs; Entrepreneur; Manager Vs. Entrepreneur.

2. Opportunity / Identification and Product Selection: Entrepreneurial Opportunity Search and Identification; Criteria to Select a Product; Conducting Feasibility Studies; Project Finalization; Sources of Information.

3. Small Enterprises and Enterprise Launching Formalities : Definition of Small Scale; Rationale; Objective; Scope; Role of SSI in Economic Development of India; SSI; Registration; NOC from Pollution Board; Machinery and Equipment Selection; Project Report Preparation; Specimen of Project Report; Project Planning and Scheduling using Networking Techniques of PERT / CPM; Methods of Project Appraisal.

4. Role of Support Institutions and Management of Small Business : Director of Industries; DIC; SIDO; SIDBI; Small Industries Development Corporation (SIDC); SISI; NSIC; NISBUD;

State Financial Corporation SIC; Marketing Management; Production Management; Finance Management; Human Resource Management; Export Marketing; Case Studies-At least 4 (four) in whole course

Trouble Shooting & Maintenance of Electronics Equipment's-II

1. TV System

- Working principle with block diagram of TV transmitter and receiver, Brief description with circuit diagram: TV Tuner, Video IF stage, Sound stage, Picture tube & its associated circuit, Synchronizing circuits, Horizontal & vertical deflection circuits, Remote control of a TV receiver, Idea of bandwidth, blanking and synchronization pulses, modulation scheme, colour transmission.
- Cable type TV system, Head end processor, Trunk & cable distribution system with block diagram, Scrambling.
- Introduction to LCD and LED TV systems, Introduction to high definition systems. Steps for Fault finding & Analysis.

2. Modern Appliances

- Working principle and block diagram of following: Microwave oven, Telephone, Fax machine, Printers, Scanners. Steps for Fault finding & Analysis.

Project Work